

Catching the Rug: Early Prediction of Fraudulent Memecoins on Solana via Machine Learning

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Abstract—The rapid proliferation of memecoins on blockchain platforms has increased the risk of fraudulent activities, particularly rug pulls. While previous studies have focused on Ethereum-based tokens, this paper shifts the spotlight to Solana, the leading blockchain for memecoins by trading volume and token count. Unlike Ethereum, where rug pulls often exploit smart contract backdoors, Solana memecoin rug pulls are predominantly driven by liquidity manipulation and social dynamics. This research pioneers large-scale rug pull early detection in the Solana ecosystem by assembling a dataset of 6.4 million tokens over 7 months. Market analysis reveals that a vast majority of these memecoins exhibit rug pull characteristics within one hour of launch, highlighting the urgency of short-horizon prediction. Despite the absence of code-level features, we demonstrate that classic machine learning models, particularly Gradient Boosting (XGBoost), achieve robust performance in detecting potential rug pulls using only the first 5 minutes of trading data. Furthermore, we evaluate cross-platform generalization between PumpFun and Raydium, revealing that multi-source data fusion significantly mitigates domain shift and improves detection reliability. This study advances the understanding of DeFi fraud on high-throughput chains and provides a practical framework for protecting investors.

Index Terms—Decentralized Exchange, Scam Detection, Blockchain, Machine Learning, Deep Learning, Solana

I. INTRODUCTION

The rapid advancement of blockchain technology, Web3 innovations, and decentralized finance (DeFi) has significantly fueled the expansion of the cryptocurrency market [1]. By May 2024, the market capitalization of cryptocurrencies surpassed 2 trillion US dollars [2], highlighting the profound influence of these technologies. To support token transactions and provide alternatives to peer-to-peer exchanges, various trading platforms and cryptocurrency exchanges have emerged. These platforms are categorized into centralized exchanges (CEXs) [3], which operate similarly to traditional financial institutions, and decentralized exchanges (DEXs) [4], which utilize smart contracts and cryptographic methods to ensure asset security.

The rise in DEX activity has significantly impacted the memecoin market. In 2024, memecoins emerged as a leading trend in the crypto world, not generating the largest trade

volume overall, but driving substantial activity on DEXs [5]. Memecoins are a type of crypto asset often created as a joke or to capitalize on memes and trends on the Internet [6]. Unlike traditional cryptocurrencies such as Bitcoin or Ethereum, which have specific use cases and underlying technologies, memecoins typically lack inherent utility or serious purpose. Instead, they gain popularity through social networks, community engagement, and viral marketing. Modern memecoins are tokens deployed on established blockchains (e.g., Solana, Ethereum) rather than native coins with dedicated consensus layers. However, this growth has led to an increase in fraudulent schemes, such as rug pulls [7], [8], where developers promote a token to attract investors and then withdraw liquidity, causing the token’s value to plummet. The decentralized nature of DEXs and user anonymity further complicate the detection of such schemes. As DeFi continues to expand, developing effective methods to detect rug pulls on Solana and other blockchains is crucial to safeguard users and maintain trust in the ecosystem.

Rug pulls on decentralized exchanges pose a significant threat to investor confidence and market stability. Currently, there is a lack of specialized tools for real-time rug pull detection on DEXs, allowing fraudsters to exploit this vulnerability. Consequently, there is an urgent need for mechanisms capable of identifying potential rug pulls before they occur [9], [10].

Although artificial intelligence has become a cornerstone of financial fraud detection [11], rug pulls in decentralized memecoin markets present unique challenges: extreme class imbalance, rapid token lifecycles, and chain-specific microstructures not addressed by traditional frameworks. Recent work has begun to address these challenges on specific chains: Hu et al. [12] introduced *MemeTrans*, a 40k Solana memecoin dataset focused on launchpad-phase risk annotation, while Yaremus et al. [13] developed a TVL/Idle-based detection framework for the TON blockchain. Our work extends this line of research by scaling to 6.4M Solana tokens over 7 months, emphasizing early prediction and evaluating cross-platform generalization between PumpFun and Raydium.

The main contributions of this paper are summarized as follows:

- We assemble and analyze a comprehensive dataset of 6.4 million Solana memecoins over a 7-month period, exceeding prior Solana-specific studies by over 150× in scale, which enables robust statistical analysis of rare rug pull patterns.
- We conduct a rigorous evaluation of model transferability between Solana’s primary memecoin platforms, PumpFun and Raydium, revealing significant distribution shifts and demonstrating that multi-source data fusion combined with tree-based models yields the most robust cross-platform generalization.
- We validate the efficacy of short-horizon early detection, using the first 5 minutes of trading data to forecast 1-hour outcomes, specifically within the high-throughput, liquidity-driven microstructure of Solana, showing that reliable predictions can be achieved using only tabular liquidity and trading dynamics without relying on smart contract code analysis.

II. BACKGROUND: SOLANA VS ETHEREUM MEMECOINS

Memecoins have emerged as a significant segment within the cryptocurrency landscape, characterized by their blend of humor, speculative appeal, and community engagement [14]. This section explores the contrasting community and technological ecosystems of Solana and Ethereum, two prominent blockchain platforms that support memecoins.

Solana memecoins inherently benefit from the high throughput and low transaction costs of the platform, attracting traders interested in rapid and cost-effective transactions [15]. In contrast, Ethereum memecoins leverage the platform’s established network effects and extensive user base. The Ethereum blockchain [16], which has transitioned to a proof-of-stake consensus and adopted a modular architecture, leverages established network effects. Although this approach offers flexibility and independent upgradability, it introduces complexity in layer coordination, potentially affecting user experience.

A critical issue in both ecosystems is the prevalence of rug pulls [9], a type of fraud in which developers abandon a project after securing investor funds. Ethereum’s robust DeFi infrastructure and the ease of launching tokens via the ERC-20 standard [17], [18] have made it a prime target for such scams. Ethereum memecoins, such as Shiba Inu (SHIB) and Pepe (PEPE), have gained significant traction due to community support and strategic marketing. Although smart contracts on Ethereum, primarily written in Solidity, enable decentralized, trustless, and transparent transactions, they require robust security measures such as whitelisting to manage access and reduce fraud risks.

Blockchain technology has introduced easy-to-use auditability into digital systems [19]. However, to fully benefit from this, the source code of smart contracts should be open source. While most DeFi applications adhere to this principle, many memecoins on Ethereum do not. Although the bytecode of the smart contract is always publicly available and can often

be reverse-engineered [20], [21], researchers sometimes fail to find the source code in certain notable scam cases, casting a shadow on the Ethereum memecoin ecosystem.

```

467 function fullWhitelist(address target) public
    onlyOwner {
468     authorizations[target] = true;
469     isFeeExempt[target] = true;
470     isTxLimitExempt[target] = true;
471     isInternal[target] = true;
472 }

```

Fig. 1: Ichimoku Inu’s fullWhitelist function

Most of Ethereum memecoins are based on the ERC-20 token standard, though the code may have modifications. A notable example is the inclusion of whitelists. Whitelisting in smart contracts controls access to comply with regulatory requirements such as Know Your Customer (KYC) and Anti-Money Laundering (AML) regulations [22]–[24], essential for token sales and NFT minting. Implementing whitelisting involves using Solidity’s mapping to store approved addresses and restrict access to specific functions. This ensures proper access control within the contract, providing a layer of security against potential fraud.

```

336 function _transferFrom(address sender, address
    recipient, uint256 amount) internal returns (
    bool) {
337     if (inSwapAndLiquify) {
338         return _basicTransfer(sender, recipient,
            amount);
339     }
340     if (!authorizations[sender] && !authorizations[
        recipient]) {
341         require(tradingOpen, "");
342     }
343
344     require(amount <= _maxTxAmount ||
        isTxLimitExempt[sender], "TX Limit");
345     if (isPair[recipient] && !inSwapAndLiquify &&
        swapAndLiquifyEnabled && _balances[address(
            this)] >= swapThreshold) {
346         marketingAndLiquidity();
347     }
348     if (!launched() && isPair[recipient]) {
349         require(_balances[sender] > 0, "");
350         launch();
351     }
352
353     // Exchange tokens
354     _balances[sender] = _balances[sender].sub(amount,
        "", "");
355
356     if (!isTxLimitExempt[recipient] &&
        restrictWhales) {
357         require(_balances[recipient].add(amount) <=
            _walletMax, "");
358     }
359
360     uint256 finalAmount = !isFeeExempt[sender] && !
        isFeeExempt[recipient] ? extractFee(sender,
            recipient, amount) : amount;
361     _balances[recipient] = _balances[recipient].add(
        finalAmount);
362
363     emit Transfer(sender, recipient, finalAmount);
364     return true;
365 }

```

Fig. 2: Ichimoku Inu’s _transferFrom function

Conversely, whitelist scam tokens are designed with mali-

cious intent, incorporating mechanisms to selectively restrict or enable trading for certain addresses [25]. Unlike legitimate tokens applying uniform rules, these scams use hidden mappings to maintain a whitelist of privileged addresses controlled by developers. These addresses are exempt from restrictions, allowing scammers to trade freely while non-whitelisted participants face barriers such as blocked transactions or excessive fees. This undermines the principles of transparency and fairness in decentralized finance.

The difference between a standard Ethereum memecoin such as Pepe Token [26] and a whitelist scam token such as Ichimoku Inu [27] lies in their smart contract logic. Pepe Token applies uniform trading rules, whereas Ichimoku Inu’s dynamic whitelisting (Figures 1 and 2) allows developers to manipulate trading conditions. Trading can be disabled for non-whitelisted addresses when `tradingOpen` is set to false. Whitelisted addresses are exempt from transaction limits (`isTxLimitExempt`), while non-whitelisted users remain subject to whale restrictions (`restrictWhales`). This creates unfair trading conditions that favor privileged addresses. Solana uses the Solana Program Library (SPL) as its standard for creating tokens, providing a foundation for fungible and non-fungible tokens [28]. This modular framework reduces development effort and enhances security, making scams like whitelisting abuse less feasible.

While both Solana and Ethereum offer fertile grounds for memecoin development, their distinct technological frameworks and community dynamics shape their respective ecosystems.

III. RELATED WORK

The application of machine learning to the detection of financial fraud has been extensively surveyed [11], covering tree-based models, deep learning, and graph methods in traditional domains such as credit card fraud, loan fraud, and anti-money laundering. However, rug pulls in decentralized memecoin markets present unique challenges: extreme class imbalance, rapid token lifecycles, and chain-specific microstructures not fully addressed by traditional fraud detection frameworks.

The cryptocurrency landscape has witnessed significant developments over recent years, characterized by both innovation and deception. Fraudulent activities such as Ponzi schemes [29]–[31], pump-and-dump tactics [32], [33], and rug pulls [9], [10], [13] have become increasingly prevalent. The decentralized nature of the market facilitates these scams, with rug pulls particularly leaving investors with worthless tokens after developers withdraw liquidity. The anonymity and lack of regulation [34], [35] further complicate detection and prevention.

During the initial coin offering (ICO) era [36], a surge in fraudulent schemes emerged, reminiscent of the current memecoin trend. ICOs analysis has highlighted vulnerabilities that persist in the cryptocurrency sphere [37]. These insights are especially relevant as the modern memecoin frenzy, particularly on platforms such as Uniswap V2 [38], [39], has increased rug

pull risk. Memecoins [5], [6], with their viral marketing and speculative appeal, are prime targets for fraudulent activities.

Recent advances in cryptocurrency fraud detection have focused on identifying rug pulls, especially on the Ethereum blockchain [16]. Mazorra et al. [9] developed a comprehensive labeled dataset of 26,957 tokens identified as scams on Uniswap (May 2020–September 2021), providing a theoretical classification of rug pull types—simple, sell, and trap-door—and introducing tools to identify them. Their work emphasizes token distribution metrics, such as the Herfindahl–Hirschman Index, uncovering that 90% of tokens using lock contracts like Unicrypt eventually become malicious. Srifa et al. [10] extended this to Uniswap V3, analyzing 7,450 tokens (3,212 normal, 581 rug pulls) and shifting focus from detection to timing prediction using time-series features like token volume and transaction frequency.

Beyond liquidity-based signals, behavioral pattern profiling has been explored for early warning. Cao et al. [40] constructed 12 token-level features based on three wash-trading patterns (Self, Matched, Circular) for BSC meme tokens, achieving PR-AUC = 0.9185 with Random Forest. Their work emphasizes lead-time analysis (mean 3.8 hours of early warning) and interpretable feature ablation. Although their focus is BSC and wash-trade signals, our approach targets Solana and liquidity-based labels (TVL/Idle), suggesting that combining behavioral and liquidity signals could improve cross-chain detection.

Beyond feature-based machine learning, graph neural networks have shown promise in modeling token interaction patterns. Wu et al. [41] proposed RugScreener, a temporal GNN architecture that captures dynamic transaction graphs for ERC-20 tokens on Ethereum. While their approach leverages relational structure, our work focuses on interpretable, tabular features derived from liquidity dynamics and trading behavior—offering a complementary, lower-complexity alternative suitable for real-time deployment on high-throughput chains like Solana.

Beyond Ethereum, recent work has extended rug pull detection to other blockchains. Yaremus et al. [13] developed a machine learning framework for The Open Network (TON), analyzing data from STON.fi and DeDust DEXs. Like our work, they compare two rug pull definitions—TVL-based liquidity withdrawal and idle-based trading cessation—and demonstrate that Gradient Boosting models can identify scams within five minutes of trading. Their key finding that feature distributions differ significantly across DEXs directly motivates our cross-platform analysis of Pumpfun and Raydium.

While most prior rug pull detection research targets Ethereum [9], [10], Hu et al. [12] recently released MemeTrans, a dataset of 40k Solana memecoins with features spanning trading activity, holding concentration, and bundle-level entity resolution. Their annotation combines statistical indicators with the detection of manipulation-patterns, achieving a 56.1% reduction in simulated financial loss. Our work complements this by scaling to 220× more tokens (6.4M vs. 40k), using liquidity-based rug pull definitions (TVL drop /

prolonged idleness) rather than launchpad-phase heuristics, and evaluating cross-platform generalization between Pumpfun and Raydium.

IV. METHODOLOGY

A. Data Collection and Preprocessing

The data used for this research was collected directly from blockchain nodes and included transaction records, especially focusing on the minting of new tokens. Data from two decentralized exchanges (DEXes) were analyzed: Raydium and PumpFun. It is important to note that these platforms are interconnected: tokens that achieve a market capitalization of \$69,000 on PumpFun are automatically migrated to Raydium for further trading. And from November 2024 to early 2025, the crypto market witnessed Memecoin’s second explosive growth. The key event in this wave was the deep integration of AI concepts with meme culture. For each DEX, the following data columns were extracted and standardized:

PumpFun		Raydium
block_time	slot	slot
tx_idx	creator	tx_idx
name	symbol	address
url	mint	signature
bundle_size	gas_used	block_time
amount_of_instructions	amount_of_lookup_reads	signing_wallet
amount_of_lookup_writes	bundle_structure	direction
bundled_buys	bundled_buys_count	base_coin
dev_balance	creation_ix_index	quote_coin
curve_address	pf_program_index	base_coin_amount
direct_pf_invocation	version	quote_coin_amount
mayhem_mode	token_program	base_pool_before
signature	parent_program	quote_pool_before
signing_wallet	direction	base_pool_after
base_coin	base_coin_amount	quote_pool_after
quote_coin_amount	virtual_token_balance_after	serum_market_id
virtual_sol_balance_after	provided_gas_fee	raydium_market_id
provided_gas_limit	fee	provided_gas_fee
consumed_gas		provided_gas_limit
		fee
		consumed_gas
		parent_program

TABLE I: Data columns extracted from Raydium and PumpFun.

The data was collected over a 7-month period for each DEX (Nov. 30, 2024 – Jun. 30, 2025). Given the high transaction activity on the Solana blockchain, this timeframe provided sufficient observations for robust modeling. After obtaining the data via API, we performed preprocessing to construct a dataset of memecoin-SOL trading pairs. Table II demonstrates the scale of our token dataset and the wealth of transaction records.

DEX	Number of tokens	Total Swap Transactions
Raydium	97965	5.923951×10^8
PumpFun	6304235	4.226951×10^8

TABLE II: Dataset statistics for Raydium and PumpFun.

To effectively model token behavior, additional features were derived from the raw data, capturing both token performance on the DEXs and token creator-specific attributes.

These features were aggregated into a structured dataset to serve as input to the machine learning model. Table III lists the basic data features required for this study.

Feature	Description
count_tx	Total number of transactions associated with the token.
purchase_percentage	Percentage of transactions classified as token purchases.
sale_percentage	Percentage of transactions classified as token sales.
unique_buyers	Number of unique wallets that purchased the token.
unique_sellers	Number of unique wallets that sold the token.
total_sol_value	Total value of transactions in SOL associated with the token.
sell_sol_value	Total SOL value from token sale transactions.
buy_sol_value	Total SOL value from token purchase transactions.
buy_sell_cnt_ratio	Ratio of buy transaction count to sell transaction count.
buy_sell_value_ratio	Ratio of buy SOL value to sell SOL value.
price_change_first_to_3_blocks	Price change of the token during its first three blocks of activity.
price_change_first_to_last	Price change from first to last trade.
buy_price_std	Standard deviation of buy prices.
sell_price_std	Standard deviation of sell prices.
first_buy_time	Timestamp of the first buy transaction.
first_sell_time	Timestamp of the first sell transaction.
min_pool_info_time	Timestamp of the earliest pool information.
max_pool_info_time	Timestamp of the latest pool information.
last_trade_time	Timestamp of the last trade.
max_price	Maximum price.
min_price	Minimum price.
min_price_after_max	Minimum price after the maximum price.
first_price	Price at the first trade after mint.

TABLE III: Calculated features to observe token activity.

B. Rug Pull Target Variable Definition

Following prior work on multi-DEX rug pull detection [13], this study adopts both Idle and TVL approaches [42], [43] to define rug pulls, each corresponding to a distinct failure mode in decentralized markets.

TVL Approach: The Total Value Locked is an important investment indicator. It is typically the total value of the LPs over a certain period of time. The higher the total value, the stronger its liquidity.

$$TVL_t = \sum_{p \in \mathcal{P}} \left(P_{p,t}^{(base)} \cdot Q_{p,t}^{(base)} + P_{p,t}^{(quote)} \cdot Q_{p,t}^{(quote)} \right)$$

TVL rug pulls are one of the most common scams in meme investing. They typically use false advertising, promises of high returns, and marketing pump-and-dump schemes to attract users to deposit assets. At this point, TVL can experience a rapid increase in value in a short period, followed by an immediate withdrawal of funds from the liquidity pool, causing TVL to drop by 99%.

Idle Approach: The idle approach defines a rug pull in which no transactions occur for an extended period:

$$Idle_t = \mathbb{I} \left(\sum_{\tau=t-w}^t V_\tau = 0 \right).$$

When a token enters a dormant state for a long time, it effectively loses liquidity. Specifically, when the inactivity time ratio is high (e.g., close to 0.5), the market exhibits low trading intensity, and thus low liquidity.

TVL OR Idle Rug Pull: Based on the above definition, when a token triggers a TVL Rug Pull condition or becomes Idle during trading, the token is classified as a high-risk token. The specific determination rules are as follows:

The maximum drawdown depth (MDD) of TVL is defined as follows: $MDD_t = \frac{TVL_t - \max_{\tau < t} TVL_\tau}{\max_{\tau < t} TVL_\tau}$.

This value is between $[-1, 0]$. A negative value indicates that TVL has fallen from its peak, and the larger the absolute value, the more severe the liquidity withdrawal. Therefore, the Rug Pull event indicator variable is defined as follows:

$$Rug_t = \mathbb{I}((MDD_t < -\theta) \vee (Idle_t > \Delta t)).$$

The thresholds θ and Δt are empirically selected based on the distributional analysis of TVL drops and durations of inactivity.

C. Rolling Time Series Cross-Validation

The construction of training and test sets directly impacts the reliability of model evaluation. Since Rug Pull detection is a time-dependent task, random partitioning or standard K-fold cross-validation introduces look-ahead bias, exposing the model to future market information during training and leading to overly optimistic evaluation results. Therefore, this study employs forward rolling window cross-validation based on token issuance time, reserving the last time window as an independent test set.

Another reason for using time partitioning is that Rug Pull fraud exhibits concept drift characteristics, with its on-chain features constantly changing with the market. Compared to standard K-fold cross-validation, which mixes data from different time periods, a fixed-length rolling window utilizes only the most recent 3 months of data for training.

Furthermore, because Rug Pulls are a minority class event, this study implements a positive sample guarantee mechanism. When the number of positive samples in the validation set is insufficient, the validation window is automatically expanded to ensure that each validation fold contains at least 5 Rug Pull samples, thereby guaranteeing the statistical significance of the evaluation results.

This partitioning method follows the process of "historical training, future verification" to avoid data leakage and ensure that the final test set is used only for one-time evaluation, thereby improving the credibility of the experimental results.

D. Detection Model and Parameter Optimization

The primary objective of this study is to develop a detection learning model capable of predicting token behavior. We simplified the complex task of rug pull detection into a binary classification problem, predicting a token's future behavior using only the first 5 minutes of trading data. For traditional machine learning, we selected Random Forest, XGBoost, and MLP. We utilized Optuna (Bayesian search) to optimize hyperparameters across all folds, subsequently retraining the models with the average optimal parameters.

For transformer models, we choose FT-Transformer, Tab-Transformer, AutoInt. They represent three different ideas for tabular data modeling. We incorporated a cross-entropy loss function into our experiments to control the training and understanding of the model.

Then, we evaluated the model's predictive performance on different platforms (Pumpfun and Raydium) under transfer

training and fusion training to identify potential anisotropies in model performance.

E. Experimental Prediction Framework

Rug Pull is defined as a fraudulent event in which the developers of a project abandon it, leading to a rapid and substantial decline in the token's value. In this study, the target labels are binary indicators of whether a TVL rug pull or an Idle rug pull has occurred at 1 hour.

This paper constructs a basic experimental prediction framework Figure 3 for DeFi scenarios. The framework first collects unified transaction data related to PumpFun and Raydium through blockchain indexer, and performs feature engineering at the data layer to extract multi-dimensional features such as price dynamics, liquidity changes, and trading behavior. At the model layer, dedicated models for PumpFun and Raydium are designed, respectively, and a hybrid model integrating the distribution characteristics of the two types of data is further constructed to improve cross-platform generalization capabilities. In the prediction phase, new input samples are classified and discriminated, dividing tokens into potential rug pull Tokens or Good Tokens. Finally, in the evaluation module, the model performance is systematically verified using multiple indicators. The preset window of the experiment is $t = 5$ min, predicting the likelihood of a rug pull within a 1-hour horizon.

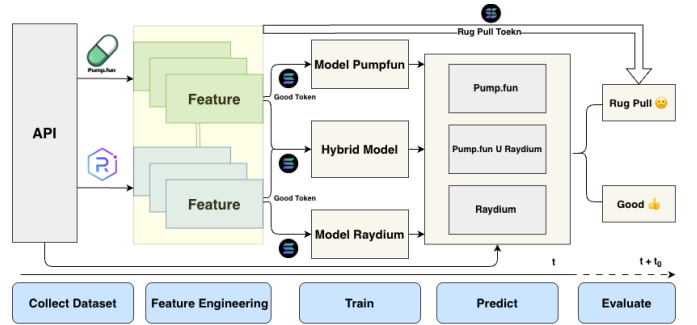


Fig. 3: The Memecoin Future Prediction Framework uses Time t to predict $t + t_0$.

F. Model Evaluation Metrics

In the crypto market, over 80% of Memecoins experience rug pulls. This research is based on unbalanced data from the real world of the cryptocurrency market. For unbalanced label recognition tasks, the accuracy (ACC) does not accurately reflect the performance of the classifier. When the data distribution is unbalanced, it often causes the output of the classifier to tend to the Rug class, which will have a higher classification accuracy, but performs poorly in the minority class Non-Rug.

To solve this problem, the study adopts evaluation indicators specifically designed for imbalanced classification. Rather than reporting all standard metrics, we focus on the three most robust indicators for detecting the minority class (Rug Pulls):

the F1-score for the positive class, the Matthews Correlation Coefficient (MCC), and the Area Under the Precision-Recall Curve (AUCPRC).

- $F1\text{-score (Positive Class)} = 2 \cdot \frac{\text{Recall} \times \text{Precision}}{\text{Recall} + \text{Precision}}$, where $\text{Recall} = \frac{TP}{TP+FN}$ and $\text{Precision} = \frac{TP}{TP+FP}$
- $MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}}$
- $AUCPRC = \text{Area under the Precision-Recall (PR) curve}$

These three metrics together provide a comprehensive and unbiased evaluation framework for the severe class imbalance present in rug pull detection.

V. NUMERICAL EXPERIMENTS

A. Data Analysis

The data contains addresses for 6.4 million Memecoins. For all Memecoins, the study collected transaction data from the first hour for preprocessing, and Table IV illustrates some basic data characteristics. This study used a data fusion method, ensuring that both the original and fused data contained valid transaction IDs.

Metric	Pumpfun (5mins)	Raydium (5mins)
Total Transactions	2.47×10^8	3.42×10^7
Avg Tx	3.93×10^1	5.88×10^2
Median Tx	1.20×10^1	3.85×10^2
Total Volume (SOL)	1.28×10^{17}	3.08×10^{16}
Avg Volume	2.04×10^{10}	5.29×10^{11}
Avg Buyers	1.83×10^1	1.54×10^2
Avg Sellers	1.30×10^1	8.42×10^1
Avg Buy Ratio	6.21×10^{-1}	7.17×10^{-1}
Avg Sell Ratio	3.94×10^{-1}	3.00×10^{-1}
Avg Price Change	1.56×10^2	5.08×10^5
Avg Buy Std	2.62×10^{-2}	1.99×10^5
Avg Sell Std	4.95×10^{-6}	6.25×10^2
Avg Last Trade time	1.62×10^2	2.97×10^2

TABLE IV: Descriptive statistics comparison between Pumpfun and Raydium.

From these characteristics, we can see that Pumpfun’s trading model is "multiple pools and small trades," while Raydium’s trading model is "few pools and large trades."

B. Rug Pull Labels Analysis

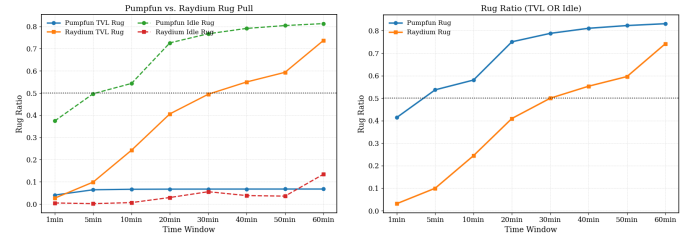
Prolonged idleness indicates that a token is not being continuously or effectively traded, while a sharp TVL drop signifies that a token lacks reliable liquidity. This study extracts rug pull labels from the feature data and performs statistical analysis to facilitate a better understanding of the machine learning models’ behavior. We define a rug pull label as a token whose TVL falls below 99% or whose idle time exceeds 80% of its lifetime without trading.

The results of this research are analyzed using these rug pull labels across different DEXs. Figure 4a and Figure 4b show that these results indicate that signals based on trading idle time are particularly effective in identifying highly speculative markets such as Pumpfun, while signals based on TVL more consistently indicate liquidity withdrawals in mature markets such as Raydium, although this indication is somewhat delayed. Overall, combining these two criteria

improves robustness and enables earlier detection of such behavior across different DEXs. When acquiring 5-minute data features, we drop data that has already been rug-pulled within that 5-min period. This reduces computational load and better reflects market trading patterns. Table V shows the distribution of the processed test dataset.

Metric	Rug Pull	Non Rug Pull
Pumpfun	43835	9711
Raydium	2931946	1893539

TABLE V: Label distribution of the testset between Pumpfun and Raydium.



(a) TVL and Idle label breakdown (b) Overall rug pull distribution

Fig. 4: Distribution of rug pull labels on Pumpfun and Raydium.

C. Detection Model Analysis

Statistical analysis of the labeled data reveals a significant upward trend in the rug pull risk of the Memecoin project over time. Specifically, the risk is relatively low in the early stages of trading, but the probability of liquidity withdrawal or price collapse gradually increases over time. Therefore, this study uses a 5-minute prediction window and a 1-hour alert window to evaluate the performance of different machine learning models in the rug pull prediction task.

Combining the experimental results in Table VI, it can be further observed that different data domains (such as Raydium, PumpFun, and their cross-DEX transfer) have a significant impact on model performance. In general, training within the same domain (such as Raydium or PumpFun) shows stable performance, with MCC scores around 0.25–0.36 across most models. However, cross-DEX prediction (such as Raydium→PumpFun) leads to a significant performance drop, with MCC values approaching zero or becoming negative across nearly all models, indicating substantial distributional differences between the two DEX platforms. Among the models, tree-based approaches (Random Forest and XGBoost) consistently outperform neural network-based models (MLP, FTTransformer, TabTransformer, and AutoInt) in cross-domain scenarios, demonstrating stronger robustness to domain shifts.

Interestingly, when models are trained on the fused dataset (Raydium U PumpFun) and evaluated on either domain, they achieve results comparable to or even better than single-domain training on most metrics. For instance, in the FusionβPumpFun direction, XGBoost and Random Forest

achieve MCC values of 0.3947 and 0.3942, respectively, which are superior to the PumpFun/PumpFun domain-specific results (0.3562 and 0.3558). This suggests that multi-source data integration effectively improves the model’s generalization ability across DEX environments. In particular, XGBoost and Random Forest exhibit the most robust performance on AUCPRC and F1 metrics when leveraging fused data, validating the effectiveness of the data fusion strategy in the rug pull prediction task. However, it should be noted that Transformer-based models (FTTransformer, TabTransformer, AutoInt) show limited improvement from fusion on the Raydium testset (MCC values of -0.0444, -0.0543, and 0.0247, respectively), indicating that advanced neural architectures do not necessarily guarantee better cross-domain transferability in this specific financial prediction context. Overall, these findings highlight the importance of domain-aware model selection and multi-source data integration for building reliable DeFi risk prediction systems. We hypothesize that Raydium’s more mature, lower-noise trading data provides a stronger foundational signal of legitimate liquidity, which helps the model generalize better to the noisier PumpFun environment when fused.

TABLE VI: Rug Pull Prediction Results (TVL OR Idle). TVL OR Idle denotes rug pulls identified by either 99% TVL drop OR 80% idle time. Best results per column are **bolded**.

Method	TVL OR Idle						
	R	R→P	P	P→R	R ∪ P	F→P	F→R
Random Forest							
F1(1)	0.7524	0.7207	0.7785	0.6106	0.7838	0.7877	0.7408
MCC	0.2874	-0.0434	0.3558	-0.2433	0.3938	0.3942	0.2246
AUCPRC	0.7578	0.5303	0.7634	0.5435	0.7897	0.7978	0.6040
XGBoost							
F1(1)	0.7532	0.7238	0.7810	0.3778	0.7833	0.7885	0.7288
MCC	0.2914	-0.0026	0.3562	-0.0664	0.3888	0.3947	0.1543
AUCPRC	0.7548	0.5771	0.7564	0.5792	0.8003	0.8011	0.6426
MLP							
F1(1)	0.7482	0.7221	0.7800	0.7068	0.7333	0.7362	0.6916
MCC	0.2663	-0.0427	0.3517	0.0865	0.3143	0.3144	-0.1121
AUCPRC	0.7379	0.5019	0.7457	0.5317	0.7437	0.7588	0.5833
FTTransformer							
F1(1)	0.7510	0.7247	0.7796	0.7130	0.7674	0.7715	0.7174
MCC	0.2807	-0.0508	0.3649	0.0387	0.3232	0.3257	-0.0444
AUCPRC	0.7506	0.4654	0.7428	0.5576	0.7297	0.7470	0.5474
TabTransformer							
F1(1)	0.6286	0.7408	0.7621	0.3530	0.7480	0.7543	0.6295
MCC	0.2508	0.0991	0.2867	-0.1479	0.2580	0.2719	-0.0543
AUCPRC	0.6997	0.6628	0.7168	0.5224	0.7003	0.7093	0.5686
AutoInt							
F1(1)	0.7501	0.7294	0.7797	0.6992	0.7694	0.7692	0.6851
MCC	0.2758	-0.0575	0.3491	-0.0946	0.3339	0.3307	0.0247
AUCPRC	0.7346	0.4775	0.7492	0.5137	0.7249	0.7354	0.5854

Note: R = Raydium (train/test on Raydium), P = PumpFun (train/test on PumpFun), F = Fusion (train on R ∪ P), → = transfer learning direction, ∪ = combined dataset.

VI. LIMITATIONS AND DISCUSSION

This study should be considered as a baseline exploration of rug pull prediction in decentralized exchanges. Although several models achieve relatively strong performance (e.g.,

AUCPRC exceeding 0.8 in certain settings), these results are not yet sufficient for real-world deployment, especially in high-risk financial environments where false negatives carry substantial investor losses.

Our results reveal substantial performance degradation under cross-DeFi settings (Table VI), indicating distribution shift between platforms. This aligns with the findings of Yaremus et al. on TON [13], where cross-DEX transfer learning also suffered from feature distribution mismatch. The pattern suggests that platform-aware adaptation—rather than naive data fusion—may be necessary for robust cross-DEX rug pull detection.

Our dataset (6.4M tokens) substantially exceeds MemeTrans [12] (40k tokens), allowing a more robust statistical analysis of rare rug pull patterns. However, MemeTrans provides richer bundle-level features for entity resolution, suggesting a promising direction for future feature engineering. Integrating bundle detection could help identify coordinated manipulation that our current feature set may miss.

The current study relies on liquidity dynamics and price volatility features (Table III), reflecting Solana’s distinct market microstructure. Cao et al. [40] demonstrate that trade-level wash-trading features are primary drivers of detection performance in BSC. Future work could fuse both signal types—liquidity-based and behavioral—for more robust cross-chain generalization.

We employ standard tabular models, which may not fully capture the temporal dynamics, liquidity evolution, and microstructure patterns inherent in DeFi markets. More advanced modeling techniques—such as temporal GNNs [41] or sequence-based transformer architectures—could better represent these complex dynamics, albeit at a higher computational cost.

VII. CONCLUSION AND FUTURE WORK

This paper presents the first large-scale systematic study of rug pull prediction on Solana memecoins, analyzing 6.4M tokens. We assemble the largest Solana memecoin dataset to date, exceeding prior work by 200× [12]. We implement and compare two rug pull definitions—TVL-based liquidity withdrawal and Idle-based trading cessation—within a unified study, following the methodology validated on TON [13]. We evaluate the generalization of the model between Pump.fun and Raydium, revealing a significant distribution shift that challenges naive data fusion approaches. We demonstrate that Gradient Boosting models can identify rug pulls within 5 minutes of trading, with XGBoost achieving the highest cross-platform generalization on fused data.

The experimental results highlight that different platforms exhibit distinct trading characteristics: PumpFun is characterized by high-frequency, small-volume trades, whereas Raydium tends to involve lower-frequency, large-volume trades. These structural differences lead to varying data distributions and directly affect model performance.

Data fusion proved feasible and promising, as the combination of data from multiple platforms improved model robustness and partially mitigated domain-specific biases. This

suggests the potential for developing more generalizable cross-platform prediction models.

For future work, several directions are worth exploring. First, advanced architectures such as temporal GNNs [41] could be introduced to capture dynamic transaction patterns. Second, feature enrichment could integrate bundle-level entity resolution [12] and wash-trading patterns [40] with our liquidity-based features. Third, shorter prediction horizons below 5 minutes could be investigated for earlier warning, balancing accuracy against actionable lead time. Finally, real-world deployment could involve collaboration with DEX platforms to validate detection systems in production environments and measure actual investor loss prevention.

As Solana continues to dominate the memecoin market [5], robust fraud detection mechanisms become increasingly vital. This work provides a foundation for protecting investors and maintaining trust in the Solana DeFi ecosystem.

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